



Bacterial Diarrhea — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. By the WHO definition, diarrhea is:

EASY

- A. One loose stool a day
- B. Two formed stools a day
- C. Any stool with mucus
- D. Three or more loose or liquid stools per day
- E. Stool lasting one week

2. Which agar shows *E. coli* O157:H7 as colourless colonies?

EASY

- A. EMB agar
- B. Hektoen Enteric agar
- C. Skirrow agar
- D. TSI agar
- E. Sorbitol MacConkey agar

3. Which organism is comma- or S-shaped and microaerophilic?

EASY

- A. *Shigella sonnei*
- B. *Salmonella typhimurium*
- C. *E. coli*
- D. *V. cholerae*
- E. *Campylobacter jejuni*

4. Which *E. coli* pathotype is the most common cause of travellers' diarrhea?

EASY

- A. EPEC
- B. STEC
- C. EIEC
- D. EAEC
- E. ETEC

5. A bacterium gives an alkaline slant, acid butt, no gas and no H₂S on TSI. It is non-motile and non-lactose fermenting. Which?

MEDIUM

- A. *Salmonella*
- B. *Shigella*
- C. *E. coli*
- D. *Campylobacter*
- E. *Vibrio cholerae*

6. Why do *Salmonella* colonies appear black on Hektoen Enteric agar?

MEDIUM

- A. Lactose fermentation
- B. Sorbitol fermentation
- C. H₂S production with ferric ammonium citrate
- D. Gas production
- E. Oxidase activity

7. Which organism has the lowest infective dose (~10² organisms)?

MEDIUM

- A. *Salmonella*
- B. *Vibrio cholerae*
- C. Enterotoxigenic *E. coli*
- D. *Shigella*
- E. None; all need ≥10⁶

8. A methylene-blue stool smear shows many neutrophils. Which organism is most likely?

MEDIUM

- A. *V. cholerae*
- B. Enterotoxigenic *E. coli*
- C. *Clostridium perfringens*
- D. *Shigella*
- E. *Staphylococcus aureus*

9. A 3-year-old with non-bloody *Salmonella* enterocolitis is otherwise well. What is the best approach?

MEDIUM

- A. Ciprofloxacin to shorten illness
- B. Antimotility drugs
- C. Fluids; antibiotics do not shorten illness
- D. Gentamicin injection
- E. Surgery

10. For uncomplicated shigellosis, which is recommended?

MEDIUM

- A. Loperamide
- B. Prolonged corticosteroids
- C. Oral fluids; avoid antidiarrhoeals
- D. Immediate colectomy
- E. Only IV antibiotics

11. A 6-year-old has bloody diarrhea then acute renal failure after eating a burger. Which pathotype?

MEDIUM

- A. ETEC
- B. STEC (O157:H7)
- C. EPEC
- D. EAEC
- E. EIEC

12. A traveller has dysentery; the isolate is non-motile and a non-/late-lactose fermenter, but it is *E. coli*. Which pathotype?

MEDIUM

- A. EIEC
- B. ETEC
- C. EPEC
- D. STEC
- E. EAEC

13. Heat-stable enterotoxin of ETEC increases which intracellular messenger?

MEDIUM

- A. cAMP
- B. cGMP
- C. IP₃
- D. Calcium only
- E. ATP

14. Which conditions suit isolation of *C. jejuni*?

MEDIUM

- A. Blood agar at 25 °C aerobically
- B. MacConkey at 37 °C
- C. Sorbitol MacConkey
- D. EMB at 42 °C aerobically
- E. Skirrow agar at 42 °C in 5% O₂ + 10% CO₂

15. A patient develops ascending paralysis after a diarrheal illness from poultry. What mechanism links them?

HARD

- A. Shiga toxin action on nerve cells
- B. LT toxin activation of cAMP
- C. Direct CNS invasion by *Shigella*
- D. Molecular mimicry between *C. jejuni* surface polysaccharide and nerve glycoprotein sugars
- E. Endotoxin shock

16. Both are non-lactose fermenters. Which set separates Salmonella from Shigella?

HARD

- A. Salmonella is motile and H₂S-positive
- B. Salmonella is non-motile and H₂S-negative
- C. Salmonella needs microaerophilic growth
- D. Shigella is H₂S-positive
- E. Both are identical on TSI

17. A patient has fever, diarrhea and a positive blood culture in the first week. Which sample is most likely to reveal the organism in enteric fever?

HARD

- A. Blood culture
- B. Stool culture only
- C. Urine only
- D. Sputum
- E. Skin swab

18. A patient on a PPI has a higher risk of Salmonella infection at the same exposure. Why?

HARD

- A. Reduced gastric acid lowers the protective barrier
- B. PPIs increase salivary IgA
- C. PPIs raise peristalsis
- D. PPIs kill normal flora only
- E. PPIs make bacteria motile

19. A child passes infrequent, large-volume watery stools without blood or fever. Which mechanism is most likely?

HARD

- A. Invasion and destruction of mucosa
- B. Toxin-induced fluid secretion overwhelming colonic absorption
- C. Microabscess formation in the ileum
- D. Pseudomembrane formation
- E. Cell-to-cell spread

20. Why do *E. coli* enterotoxins cause small-bowel rather than colonic disease?

HARD

- A. The colon is acidic
- B. The colon has more IgA
- C. Colonic cells probably lack the toxin receptors
- D. Pili adhere only in the colon
- E. Toxins are inactivated by bile only

✓ Answer key

Q	Answer	Difficulty	Why
1	D	EASY	WHO: ≥ 3 loose stools per day or more than usual for that person.
2	E	EASY	O157 does not ferment sorbitol.
3	E	EASY	C. jejuni grows best in 5% O ₂ and at 42 °C.
4	E	EASY	ETEC produces ST and LT enterotoxins.
5	B	MEDIUM	Shigella makes neither gas nor H ₂ S and is non-motile.
6	C	MEDIUM	H ₂ S plus ferric ammonium citrate gives black colonies.
7	D	MEDIUM	Salmonella and vibrios need 10 ⁵ –10 ⁸ . (The epidemiology slide says "as few as 10"; the pathogenesis slide says ~10 ² .)
8	D	MEDIUM	Neutrophils indicate invasion; toxin producers do not cause this.
9	C	MEDIUM	Antibiotics may prolong excretion, raise carrier state and select resistance.
10	C	MEDIUM	Antidiarrhoeal drugs may worsen Shigella dysentery.
11	B	MEDIUM	Shiga toxin causes haemorrhagic colitis and HUS.
12	A	MEDIUM	EIEC invades epithelium and mimics shigellosis.
13	B	MEDIUM	ST raises cGMP; LT resembles cholera toxin and raises cAMP.
14	E	MEDIUM	Growth is at 42 °C in a microaerophilic atmosphere; no growth at 25 °C.
15	D	HARD	Guillain-Barré can follow Campylobacter enteritis through molecular mimicry.
16	A	HARD	Shigella: non-motile, no H ₂ S, no gas; Salmonella: motile, H ₂ S+.
17	A	HARD	In enteric fever a blood culture is most likely to be positive in the first 2 weeks; stool suits enterocolitis.
18	A	HARD	Gastric pH is a key defence; acid-lowering drugs reduce it.
19	B	HARD	Secretory diarrhea produces infrequent large-volume stools; dysenteric illness causes frequent bloody stools.
20	C	HARD	Toxins are cell-specific; the colon is not susceptible, probably lacking receptors.